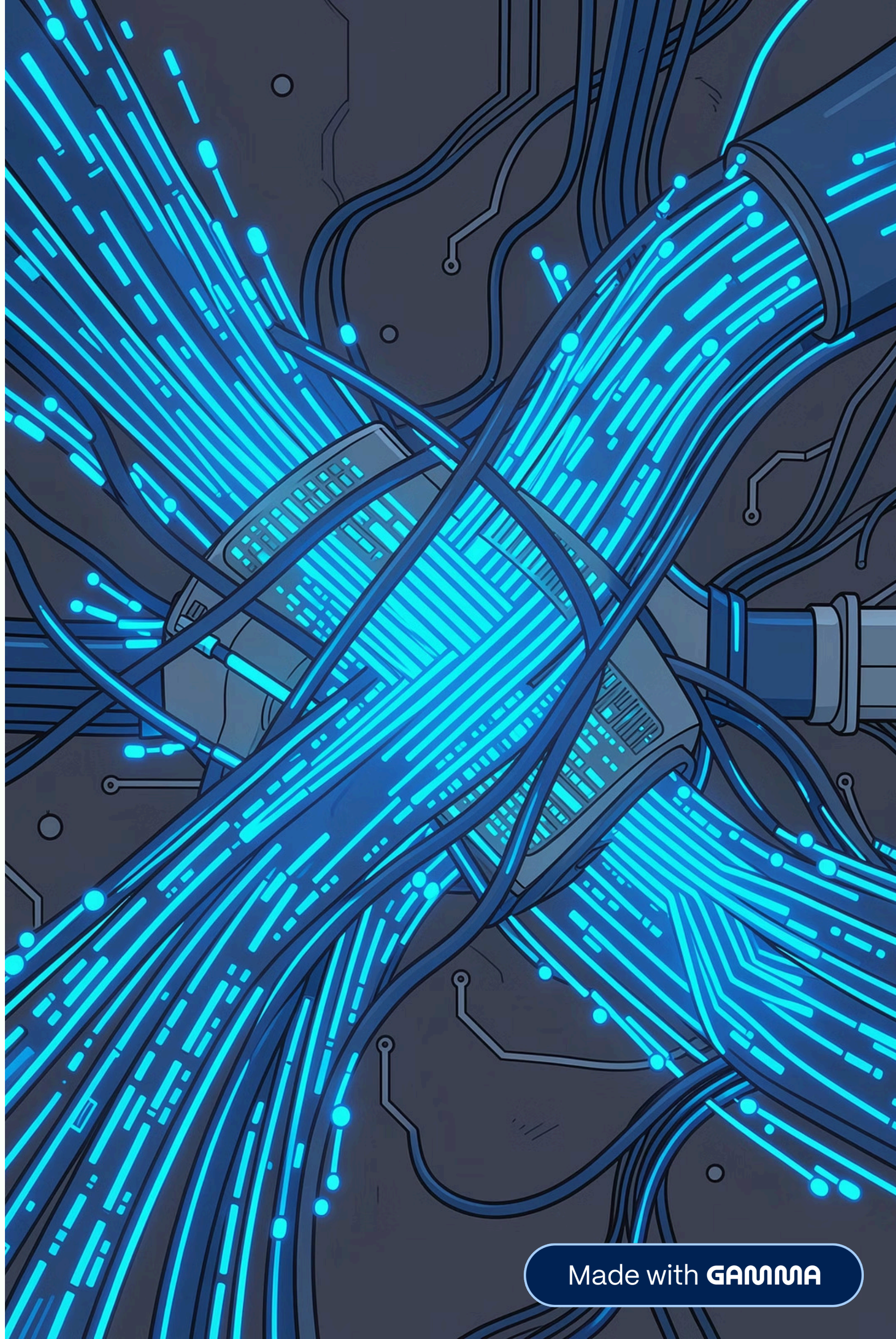
Telco Customer Churn Analysis

Business Analytics Case Study | Excel-Based Analysis

GIFT · ANALYSTLAB AFRICA INTERNSHIP · JUNE 2026



The Scale of the Problem

The company is losing **more than 1 in 4 customers** – a structural challenge with clear, addressable drivers.

7,043

Total Customers

26.5%

Churn Rate

1,869 churned customers

$139K

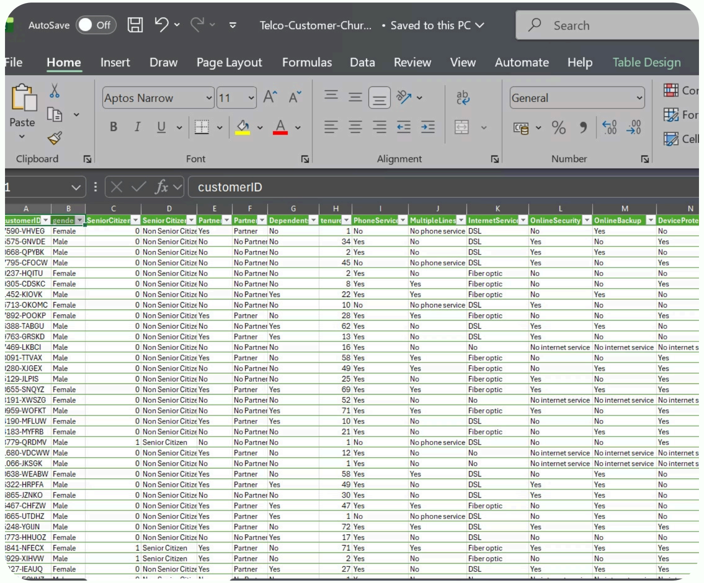
Monthly Revenue at Risk

5,174

Retained Customers

⚠ Without intervention, the business risks continued revenue erosion, higher acquisition costs, and deteriorating customer lifetime value.

Dataset Overview



7,043 customer records across four data categories, analyzed entirely in Microsoft Excel.

Demographics

Gender, Senior Citizen, Partner, Dependents

Account Details

Tenure, Contract Type, Payment Method

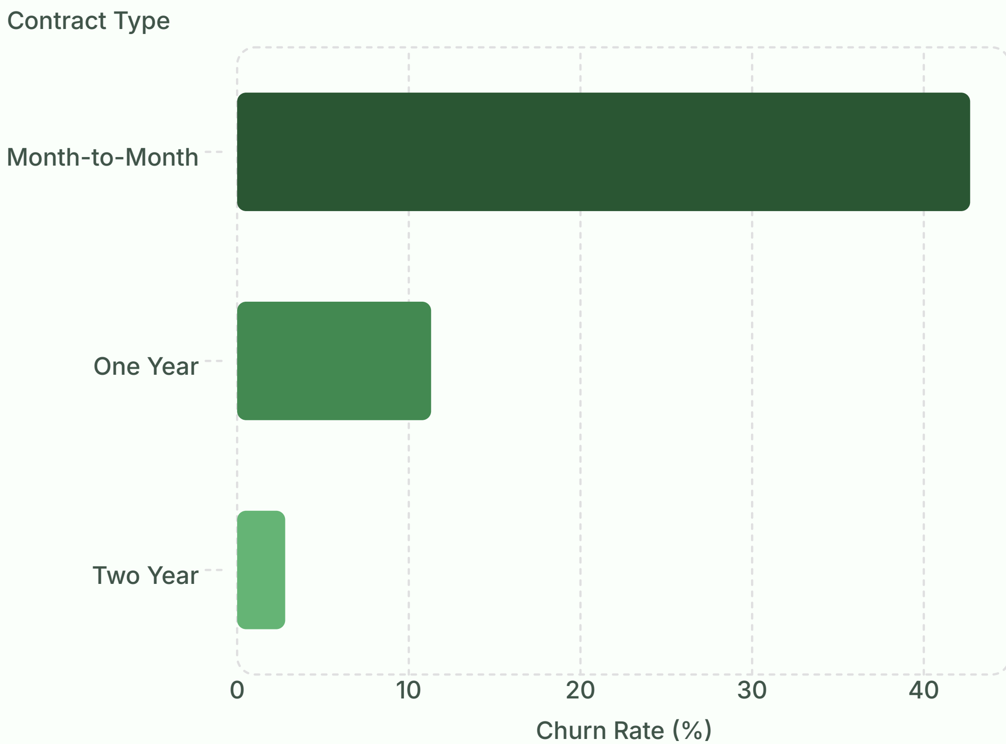
Services Used

Phone, Internet, Streaming, Security, Tech Support

Financials

Monthly Charges, Total Charges

Top Finding: Contract Type Is the Strongest Predictor



15× Higher Churn

Month-to-month customers churn at **42.7%** – 15 times the rate of two-year contract holders. This pattern holds across every dimension analyzed.

~1,655

Month-to-month churned

~166

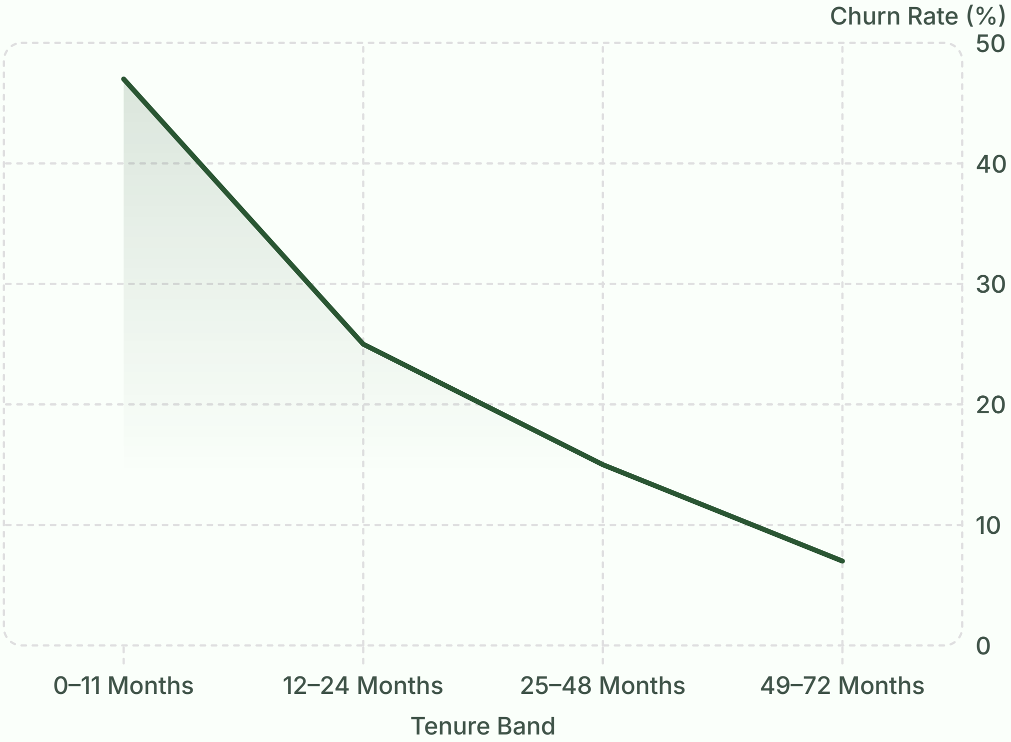
One-year churned

~48

Two-year churned

Tenure & Internet Service: Critical Risk Zones

Tenure Churn Trend



Key Insights

Early months are critical: Churn exceeds 47% in the first year, falling below 7% for customers with 60 - 72 months tenure. Churned customers average only 18 months vs. 37.7 for retained.

Fiber optic paradox: Despite being the premium product, fiber optic customers churn at **41.9%** – more than twice the DSL rate (19.0%).

Demographics & Payment Behavior

Senior Citizens

41.7% churn rate vs. 23.7% for non-seniors – potentially reflecting affordability or service complexity concerns.

Customers Without Partner

33.0% churn rate vs. 19.7% for those with a partner. Without dependents: 31.3% vs. 15.5%.

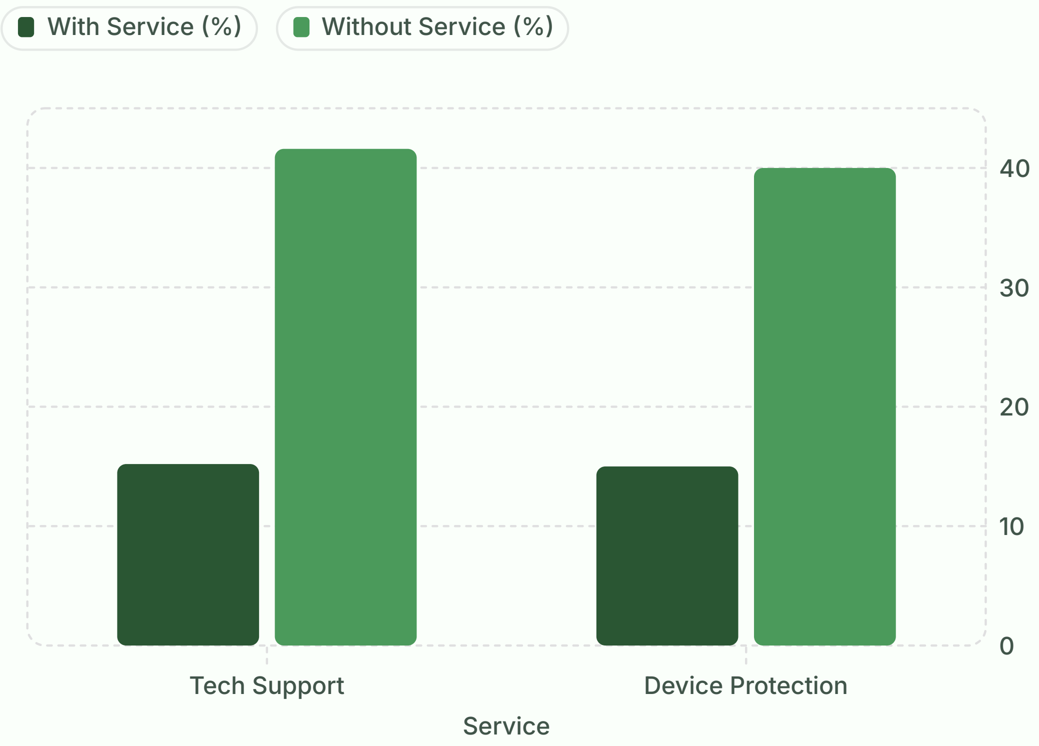
Electronic Check Payers

45.3% churn rate – the highest of any payment method. 1,071 churned customers paid this way vs. ~15-16% for auto-pay users.

Paperless Billing

33.6% churn rate – double the 16.4% rate of paper billing customers, likely overlapping with the high-risk month-to-month segment.

Add-On Services: The Loyalty Anchor



Add-On Services Reduce Churn by ~65%

Customers subscribed to Tech Support churn at just **15.2%** vs. 41.6% without it. Device Protection shows a similar pattern.

These services act as **loyalty anchors** – bundling them more aggressively could significantly lift overall retention.

Cross-selling add-ons during onboarding and renewal is one of the highest-ROI interventions available.

Business Insights Summary

Patterns Discovered

- Churn is **structural, not random** – contract type, tenure, and service type create predictable risk tiers.
- Risk **declines linearly with time** – every additional year roughly halves churn probability.
- Auto-pay users churn at **one-third the rate** of manual payers – a reliable commitment proxy.

Highest-Risk Segments

Month-to-Month

42.7% churn – largest volume

New Customers (0–12 mo)

47.7% churn – onboarding gap

Fiber Optic

41.9% churn – premium product

Electronic Check

45.3% churn – highest payment risk

Five Strategic Recommendations

01

Contract Migration Incentive

Reward month-to-month customers who upgrade to annual/two-year contracts. **Expected impact: HIGH**

02

First-Year Onboarding Journey

90-day program with touchpoints at Day 7, 30, 60. Drive add-on adoption early. **Expected impact: HIGH**

03

Auto-Pay Conversion Campaign

Offer $3 - $5/month credit to switch electronic check users to auto-pay. **Expected impact: MEDIUM-HIGH**

04

Address Fiber Optic Dissatisfaction

Survey fiber customers, fix pricing or reliability issues, consider loyalty rate reductions. **Expected impact: HIGH**

05

Bundle Add-On Services

Discounted bundles with Tech Support or Device Protection; 30-day free trials for high-risk customers. **Expected impact: MEDIUM**

Conclusion & Key Takeaways

The Path Forward

Churn is not random – it follows predictable, addressable patterns. Implementing these five recommendations in sequence offers a realistic path to reducing churn by **8-12 percentage points**, recovering hundreds of thousands in monthly recurring revenue.

Key Takeaways at a Glance

- **7,043** customers analyzed · **1,869** churned (26.5%)
- **$139,131** monthly revenue at risk
- Month-to-month (**42.7%**) and first-year customers (**47.7%**) are highest-risk
- Add-on services reduce churn by up to **65%**
- Auto-pay adoption is the **simplest, fastest win** available